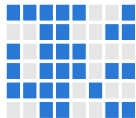
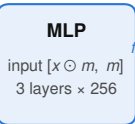


(a) Pre-training, once per dataset

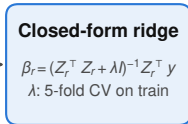
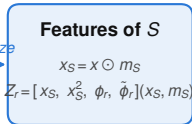
rows $\times$ all columns held,
random visibility mask m



■ visible ■ hidden



freeze



Objective

predict targets + reconstruct hidden columns; freeze last hidden layer ϕ_r ; untrained twin $\tilde{\phi}_r$ (seeds $r=1, 2, 3$)

(b) Query (S, y): subset-specific readout

Estimate

$\hat{y} = \frac{1}{3} \sum_r Z_r \beta_r$; $\hat{V}_y(S) = R_{\text{test}}^2(\hat{y}) + \text{closure}$
safeguards: s.d. floor; clip features, $\hat{y}$ to train range

(c) Scan–certify

Scan

$\hat{V}_y$ for all $|S| \leq K+1 \rightarrow$
 $\hat{M}^{(K)} + \text{ranked backgrounds}$

Certify

retrain attackers only on the top- k backgrounds $\rightarrow L^{(K)} \leq M^{(K)}$